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Statement of Purpose

I am a Master's student in Computer Science with a strong interest in interactive graphics, simulation, and machine learning. I am applying to join iLab because I am excited about the lab's interdisciplinary research at the intersection of computer vision, graphics, and human-computer interaction. My goal is to contribute to projects that explore how visual intelligence and physical simulation can be combined to create intelligent, responsive systems.

My academic and project experience spans physically based animation (such as stable fluids and particle based incompressible fluid simulations), computer graphics, and some early exposure to machine learning. I recently built a 2D fluid simulator using OpenGL and implemented methods including PIC, FLIP, and APIC. This work gave me a deeper appreciation for the challenges and possibilities in combining numerical simulation with perceptual feedback.

I am particularly drawn to iLab's work on generative and interactive models—such as movie diffusion, 3DGS maps, or vision-guided simulation—and I would love to support data collection, implementation, and testing of such systems. I'm also eager to expand my experience in PyTorch and vision transformer models, as well as collaborate with PhD students and faculty on cutting-edge research.

Ultimately, I hope to contribute to both the creative and technical aspects of the lab's projects, and I see this as an opportunity to grow into a more capable researcher and problem solver. I am open-minded, fast-learning, and committed to making meaningful contributions to iLab.

Sincerely,

Yumeng He